

DYLAN NEFF

CERN – European Organization for Nuclear Research
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EDUCATION

Ph.D. Physics, UCLA

November 2023

M.S. Physics, UCLA

December 2017

Los Angeles, CA

Advisor: Huan Huang

B.S. Physics & B.S. Mathematics, Purdue University

May 2016

West Lafayette, IN

Advisors: Virgil Barnes, Ephraim Fischbach

RESEARCH

Affiliated Researcher – The University of Manchester

2025–Present

X17 Search at n_TOF, CERN

- Searching for a hypothetical ~ 17 MeV particle (X17) via ${}^3\text{He}(n, e^+e^-){}^4\text{He}$ at the CERN n_TOF facility, using a pressurized ${}^3\text{He}$ target surrounded by four micro-TPC Micromegas tracking walls and layered scintillator calorimetry.
- Technical lead for the Dream-based data acquisition system: designed the DAQ software front end and full acquisition chain, and assembled a new DAQ rack at CEA Saclay from spare front-end boards.
- On-site contact person for the experiment's 40-day data-taking run at CERN: coordinating commissioning and flammable-gas operations, and designing the scintillator trigger logic with six CAEN N1081B programmable logic units.
- Built the operations infrastructure that runs the experiment unattended: fifteen standing monitor processes with automated alerting on beam loss and acquisition faults, a live shift dashboard with per-wall high-voltage and run-statistics views, and automated EOS transfer with disk-space triage of every recorded run.
- Established the timing and threshold calibrations underpinning the physics data: per-channel gamma-flash time constants referenced to the beam pickup, a DREAM-to-n_TOF time map resolving an apparent ± 150 ns window into per-bunch clock drift, and a pulser-based characterization of the readout window.
- Helped build the $40 \times 40 \times 3$ cm micro-TPC Micromegas detectors at CEA Saclay and characterized them on a cosmic-ray bench; characterized a μRWELL detector in the SPS H4 beamline, measuring its efficiency and charge-spreading response and identifying a non-amplifying region no operating voltage could recover.
- Performed Geant4 simulations of the full four-arm detector to establish the feasibility of a thermal-neutron-based measurement, informing a redesign of the experiment around SiPM-read plastic and liquid-scintillator calorimeter layers; extended these to optimize the thermal-gate trigger and to decompose the aluminium (n, γ) background into its double-Compton and pair-production contributions.

Postdoctoral Researcher, CEA Paris-Saclay

2023–2025

Département de Physique Nucléaire (DPhN)

sPHENIX Experiment, RHIC

- Leading the collaboration effort to measure beam luminosity in both Au+Au and pp collisions; developed a full beam-beam simulation to constrain beam parameters via dedicated scan data.
- Developed tools to assess and validate data quality for spin physics analyses in the 2024 pp run.
- Built and deployed DAQ monitoring software with a GUI that alerts shift crews to beam losses or abnormal rates, minimizing detector dead time and allowing shift crews to focus on other tasks.

- Helped re-commission the Zero Degree Calorimeters for the polarized proton run, enabling verification of spin polarization at the collision point.

Future Electron Ion Collider (EIC) Research and Development

- Responsible for testing and characterizing prototype Micromegas detectors for the planned CyMBAL barrel tracking detector in EPIC.
- Retrofitted a cosmic muon telescope with a silicon tracker for rapid QA of prototypes; developed a synchronized DAQ and configuration bookkeeping system.
- Implemented a simulation of multiple scattering to determine optimal test beam energies and configurations, guiding design of future test beam campaigns.

Graduate Researcher, UCLA

2016–2023

STAR Experiment at RHIC

- Developed a novel observable sensitive to proton clustering; this was used to search for evidence of a first-order phase transition in the QCD phase diagram.
- Analyzed heavy-ion collision data from the RHIC beam energy scan, focusing on correlation and fluctuation measurements within the STAR working group.
- Developed DAQ rate monitoring software for real-time optimization of STAR data taking.
- Designed an automated SiPM characterization system for a new calorimeter upgrade, and participated in its commissioning at a Fermilab beam test.

Early Research, Purdue & UCLA

2014–2019

- Automated silicon photomultiplier characterization for calorimeter production.
- Participated in commissioning of a STAR forward upgrade hadron calorimeter at Fermilab.
- Led a small team conducting long-term half-life measurements of Cl-36 and Si-32 to probe potential BSM decay-rate fluctuations.
- Built a temperature-controlled system to study sensitivity of NaI(Tl) detectors to temperature and magnetic fields.

TECHNICAL SKILLS

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- **Programming & Languages:** Python, C++, C, SQL, PHP, JavaScript, Matlab, PERL, Mathematica
 - **Data Analysis & Tools:** ROOT, GEANT4, Garfield++, Git, MySQL, Jupyter, LabVIEW, Pandas/Numpy/Scipy
 - **Detector & Electronics:** Micromegas, μ RWELL, silicon photomultipliers (SiPMs), scintillators, calorimeters, NIM/VME electronics, DAQ systems, FPGA trigger logic, Arduino, Raspberry Pi
 - **Computing & Simulation:** Distributed and parallel computing, Monte Carlo methods, physics simulations, GUI development, microcontrollers, numerical integration, cluster computing environments
 - **Web & Systems:** Linux server management, LAMP stack (Linux, Apache, MySQL, PHP), front-end/backend integration, REST APIs, automation scripts

TEACHING

Nuclear Physics Lab – UCLA

2019–2022

Upper division nuclear physics lab which stressed experimental data acquisition and analysis.

Modern Physics Lab – UCLA

Fall 2019

Lab course which provided a range of modern physics experiments and analysis.

Life Science Physics Lab and Discussion – UCLA

2017–2020

Series of courses geared toward life science students.

Mechanics Lab – UCLA

2016–2017

Lab course for physics and engineering majors focused on technical writing.

AWARDS

Thesis Award – BNL RHIC & AGS Users’ Meeting (link)	2024
Young Scientist Award – Quark Matter (link)	2023

SELECT PUBLICATIONS

1. D. Neff, Z. Ji, R. Esha, G. Wang, H.Z. Huang, **Methodology for analyzing proton multiplicity fluctuations with azimuthal partitions in heavy-ion collisions**, Phys. Rev. C 110, 064905 (2024)
2. STAR Collaboration, **Probing the nature of the QCD phase transition with higher-order net-proton number fluctuation and local parton density fluctuation measurements at RHIC-STAR**, EPJ Web Conf. 296, 06005 (2024)
3. STAR Collaboration, **Recent Highlights from STAR BES Phase II**, contribution to Moriond QCD 2024, e-Print: 2405.20928 [nucl-ex] (2024)
4. D. Neff, **Exploring the QCD Phase Diagram at STAR: Measurement of Proton Fluctuations in Azimuthal Partitions**, e-Print: 2312.14628 [nucl-ex] (2023)
5. V.E. Barnes, D.J. Bernstein, C.D. Bryan, N. Cinko, G.G. Deichert, J.T. Gruenwald, J.M. Heim, H.B. Kaplan, R. LaZur, D. Neff, J.M. Nistor, N. Sahelijo, E. Fischbach, **Upper Limits on Perturbations of Nuclear Decay Rates Induced by Reactor Electron Antineutrinos**, *Applied Radiation and Isotopes* 149, 182 (2019)
6. E. Fischbach, V.E. Barnes, N. Cinko, J. Heim, H.B. Kaplan, D.E. Krause, J.R. Leeman, S.A. Mathews, M.J. Muetherthies, D. Neff, M. Pattermann, **Indications of an unexpected signal associated with the GW170817 binary neutron star Inspiral**, *Astroparticle Physics* 103, 1–6 (2018)
7. STAR Collaboration, **Azimuthal partitioned proton multiplicity fluctuations in Au+Au collisions**, *in preparation*

Further STAR and sPHENIX Collaboration publications can be found on iNSPIRE HEP.